

YOURSTORY



snowflake

H2S



COCO CLI

HACKATHON

GCC Edition



Team Name : **Sushanth Reddy (solo)**

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Team Size : **1**

Problem Statement : **Predictive Maintenance and OEE Command Center**

PlantPulse — converging IT & OT on Snowflake to predict failures, automate work orders and lift OEE

Problem Brief: downtime hides in the gap between OT and IT

Discrete manufacturing and machining plants running 24×7 rotating equipment (motors, pumps, gearboxes, spindles)

12

breakdowns in 60 days

113 h

unplanned downtime

₹18.1 L

corrective maintenance
cost

78.7%

fleet OEE vs 80% target
(PUN-L1: 76.9%)

12 / 12

showed degradation 4–7
days early, no alarm fired

Today's pain points

- ! Sensor data sits in historians**
vibration, temperature, RPM and current never meet ERP/CMMS context
- ! Fixed alarms fire too late**
thresholds sit at the danger level, so there is no lead time to plan
- ! Root cause takes hours**
engineers dig through manuals and old work orders by hand
- ! Work orders are reactive**
raised after the breakdown, then wait on out-of-stock parts
- ! OEE losses aren't tied to assets**
managers can't see which machine is costing availability

Who it's for, and how PlantPulse helps

RE

Reliability engineer

ranked alerts with evidence; cited root cause in seconds

MP

Maintenance planner

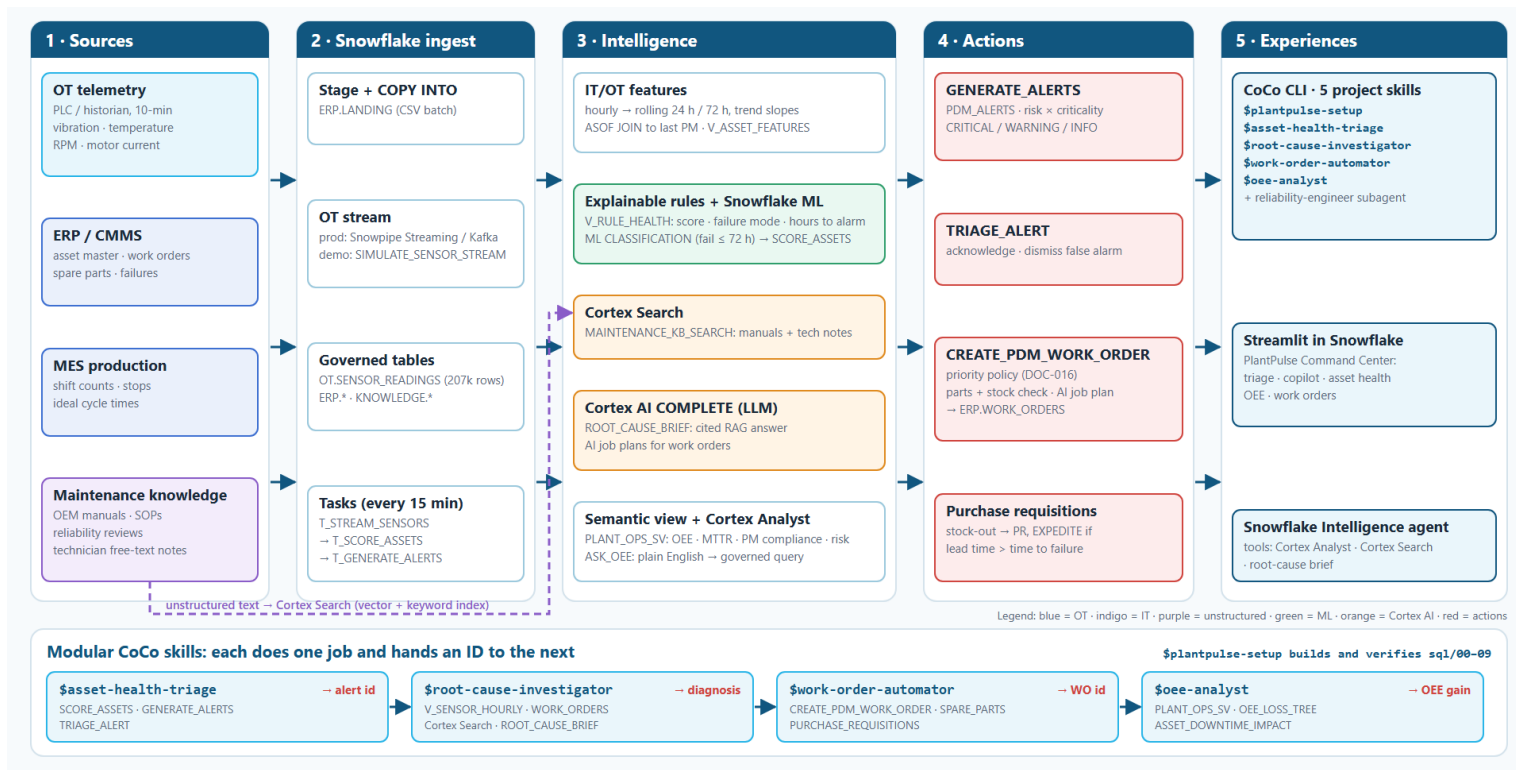
one command turns an alert into a prioritised WO with parts checked

PM

Plant manager

governed OEE, losses tied to assets, quantified gain

Architecture: IT + OT + knowledge converge in Snowflake



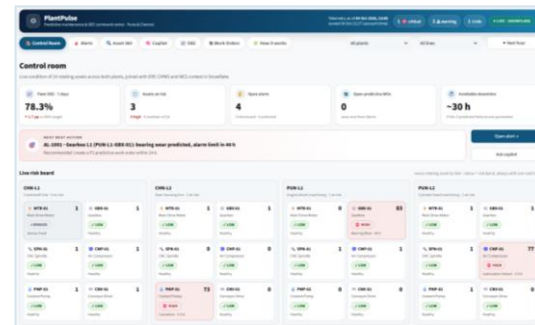
Structured: OT telemetry, ERP/CMMS, MES **Unstructured:** manuals, SOPs, technician notes **Plug-in points:** new site = data + skill · new KPI = metric

Solution: one CoCo CLI workflow from alert to work order

Five modular skills. Each queries live Snowflake objects, shows its SQL and hands an ID to the next skill

Skill	Input	Processing	Output
\$asset-health-triage	Which assets are at risk?	SCORE_ASSETS (ML + rules) · GENERATE_ALERTS	Ranked queue; AL-1001 gearbox CRITICAL; sensor fault dismissed
\$root-cause-investigator	Why is the gearbox vibrating?	Trend vs baseline · CMMS history · Cortex Search · LLM	Bearing wear recurrence, cited sources
\$work-order-automator	Create the WO for the alert	Priority policy · stock check · AI job plan	WO-PDM-90001, P2 within 24 h, parts reserved
\$oee-analyst	Worst line, and its cost?	Semantic view PLANT_OPS_SV · loss tree	PUN-L1 76.9%; +1.4 OEE pts if prevented

+ \$plantpulse-setup · reliability-engineer subagent · Snowflake Intelligence agent (Analyst + Search tools)



Command Center: 7-page Streamlit in Snowflake app

ML Classification

Cortex Search

Cortex AI

Cortex Analyst

Cortex Agent

Semantic view

ASOF JOIN

Snowpark procs

Tasks

Streamlit in Snowflake

Impact: failures caught two days early, actions in seconds

12 / 12

failures flagged in advance

48 h

median lead time
(23–69 h)

12 / 12

failure modes correctly named

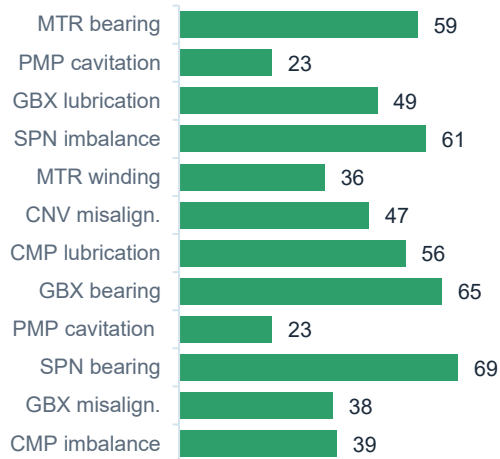
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false alarms on 60 days of history

Scales beyond the demo

Same pattern for any plant or asset type · Snowpipe Streaming for real OT · governed KPIs in one semantic view · Snowflake RBAC

Warning lead time before each failure (h)



Back-test on 60 days of synthetic history

Operational outcomes

Triage → WO

hours of manual work become < 1 min, with policy, parts and job plan

Root cause

hours of reading become ~20 s, with cited manuals and notes

Availability

+1.7 pts on PUN-L1 (≈ +1.4 OEE pts) if breakdowns become planned stops*

Spares

stock-outs raise an expedited PR at alert time, not after the failure

* Assumes a planned repair takes 35% of the breakdown duration. **Honesty note:** synthetic data. Out-of-time ML test (train before 15 Sep, test after): 4/4 held-out failures caught, median 66 h lead, hourly F1 0.92.

Additional: demo flow, roadmap and links

Demo (CoCo CLI, ~4 min)

- 1 /skill list → 5 PlantPulse skills, data landscape
- 2 \$asset-health-triage → 4 alerts ranked, dismiss sensor fault
- 3 \$root-cause-investigator → cited bearing-wear diagnosis
- 4 \$work-order-automator → WO-PDM-90001, P2, verified in ERP
- 5 \$oee-analyst → PUN-L1 losses and projected gain
- 6 Command Center → alert shows WO_CREATED

Roadmap

- CMMS write-back (SAP PM / Maximo) and a mobile technician view
- Vibration spectra / FFT features: bearing defect frequencies
- Snowpipe Streaming from the plant historian; energy KPIs
- Out-of-time ML validation as real failure history grows

Links

Code: github.com/EkaTantra/plantpulse

Live demo: ekatantra.github.io/plantpulse

In Snowflake: Streamlit app PLANTPULSE_COMMAND_CENTER

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1125

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THANK YOU

